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Explainable CNN framework with Grad-CAM analysis for real-time skin disease detection

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ABSTRACT

Skin cancer remains one of the most prevalent and potentially fatal diseases worldwide, emphasizing the need for early and reliable detection. This paper presents an explainable and lightweight convolutional neural network (CNN) model for automatic skin-lesion classification using the HAM10000 dataset. Unlike previous studies that relied on cloud computing or focused on binary melanoma detection, the proposed framework performs real-time, multi-class lesion analysis entirely on-device. The CNN model was optimized through TensorFlow Lite to achieve efficient, privacy-preserving inference on mobile platforms.

Experimental evaluation on the HAM10000 benchmark demonstrated an overall accuracy of 98.21%, confirming the model's robustness across seven lesion types. Grad-CAM and Grad-CAM++ visualization techniques were incorporated to highlight the most influential regions in each image, enhancing interpretability and clinical trust. Finally, the model was deployed within an Android mobile application enabling fully offline diagnosis and secure patient–doctor interaction. The proposed framework provides a promising step toward accessible and explainable AI-driven dermatological screening in mobile healthcare systems.

1. Introduction

Skin cancer is one of the most common cancers worldwide and remains a major public health concern. According to the World Health Organization (WHO), more than 1.5 million new cases of skin cancer are reported each year, with incidence rates continuing to rise due to increased ultraviolet (UV) exposure, aging populations, and environmental factors [1]. Early detection is essential to improving treatment outcomes and patient survival [2]. However, access to dermatological expertise is still limited in many low-resource and rural areas, resulting in delayed or inaccurate diagnoses. Conventional visual inspection by dermatologists is often subjective and affected by inter-observer variability, which highlights the need for automated, image-based diagnostic systems capable of providing consistent and reliable assessments [3,4].

Deep learning, particularly convolutional neural networks (CNNs), has shown remarkable success in visual recognition tasks and medical image analysis. In dermatology, CNN-based models have achieved dermatologist-level performance in distinguishing malignant from benign lesions. Nevertheless, most existing systems depend on cloud-based processing, which introduces latency and privacy risks, or focus solely on binary melanoma classification. These constraints limit their applicability in real-time and resource-constrained mobile healthcare environments.

Several studies have explored AI-assisted dermatology. Esteva et al. [3] trained a CNN on over 129,000 dermatoscopic images and achieved dermatologist-level performance for skin-cancer classification, while Tschandl et al. [4] developed the HAM10000 dataset—a widely adopted benchmark for skin-lesion research. More recently, explainable AI (XAI) techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) [5] have been used to visualize the image regions influencing model predictions, helping clinicians interpret AI decisions and fostering trust in automated diagnosis. However, few systems provide real-time, on-device analysis; most require continuous internet connectivity, which limits their use in low-resource areas. Mobile-based AI tools can bridge this accessibility gap by enabling rapid, offline, and cost-efficient diagnostic support while preserving patient privacy.

To address these challenges, this study proposes a lightweight and explainable CNN model for multi-class skin-lesion classification that operates entirely on mobile devices. The model is optimized using TensorFlow Lite to ensure low-latency and privacy-preserving inference. Furthermore, Grad-CAM and Grad-CAM++ techniques are integrated to

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visualize discriminative regions in the images, improving model transparency and clinical interpretability. The proposed framework is implemented within an Android application that supports offline diagnosis and facilitates patient–doctor communication.

Unlike previous works that relied on cloud-based computation or focused only on melanoma detection, the proposed system introduces a fully on-device CNN architecture capable of handling multiple skin-lesion categories in real time. The framework combines diagnostic accuracy with transparency, offering explainable and privacy-preserving inference in a mobile health (mHealth) environment. By embedding the model directly into an Android application, the solution eliminates dependency on external servers and demonstrates the feasibility of deploying explainable AI in dermatological care for resource-limited regions.

The main contributions of this study are summarized as follows:

1. Development of a lightweight CNN architecture optimized for mobile deployment and real-time inference.
2. Integration of Grad-CAM and Grad-CAM++ methods to provide visual interpretability of model decisions.
3. Implementation of a mobile dermatology application supporting offline operation and patient–doctor interaction.
4. Comparative evaluation with recent CNN-based approaches, demonstrating superior performance and explainability.

The remainder of this paper is organized as follows: Section 2 reviews related work on AI-based skin-disease detection and mobile healthcare systems. Section 3 presents the proposed CNN model, training configuration, and mobile integration. Section 4 provides experimental results and discussion. Section 5 outlines limitations and highlights future research directions. Finally, Section 6 summarizes the findings.

2. Literature review

The use of Artificial Intelligence (AI), particularly deep-learning models such as Convolutional Neural Networks (CNNs), has recently brought major progress to dermatological image analysis. Both academic research and commercial initiatives have investigated web and mobile-based approaches for automated skin-disease recognition. Nevertheless, issues related to model interpretability, device-side computational efficiency, and data privacy continue to present challenges. Advances in mobile hardware and edge-AI technologies have encouraged the development of intelligent healthcare tools capable of conducting real-time dermatological analysis directly on smartphones. The following review classifies prior work into two principal categories: non-mobile-based research and mobile-focused solutions.

2.1. Foundational research on CNNs for dermatology

The use of Convolutional Neural Networks (CNNs) has played a pivotal role in advancing dermatological image analysis over the past decade. Esteva et al. [3] conducted one of the earliest large-scale studies, training a CNN on more than 129,000 clinical images and achieving diagnostic accuracy comparable to expert dermatologists. Although the model produced remarkable results, it was primarily designed for clinical deployment and lacked integration into mobile or resource-limited environments. Brinker et al. [6] reinforced the effectiveness of CNNs through experiments on the HAM10000 dataset, demonstrating that AI systems could surpass dermatologists in melanoma-classification accuracy. This finding further validated the potential of deep learning for dermatological diagnostics. However, several early investigations, while instrumental in developing CNN modeling for skin-disease detection, did not consider mobile adaptation or lightweight deployment.

Tschandl et al. [4] introduced the HAM10000 dataset, high-quality collection of dermatoscopic images encompassing seven classes of pigmented lesions. This dataset has since become a standard benchmark for training CNNs and evaluating their performance in skin-lesion classification. Although initially created for academic research, HAM10000 has provided the foundation for numerous subsequent studies, including those later adapted for mobile frameworks. Regarding model interpretability, Selvaraju et al. [5] proposed Gradient-weighted Class Activation Mapping (Grad-CAM), an explanation method that visualizes the image regions most influential to a CNN's decision. Grad-CAM has become widely adopted in medical imaging as it enables researchers and clinicians to better understand how deep-learning models reach their predictions. While Grad-CAM can be integrated into mobile-based systems, the original study did not address on-device deployment. Furthermore, Brinker et al. [6] conducted a large-scale benchmarking analysis comparing CNN-based melanoma classifiers with the diagnostic performance of 157 dermatologists. Their findings confirmed the robustness and reliability of CNN models in identifying malignant lesions. Nevertheless, these studies collectively highlighted a key gap: the lack of mobile deployment and explainability integration within real-world dermatological diagnostic tools.

2.2. ISIC challenges and limitations

The International Skin Imaging Collaboration (ISIC) has launched several benchmark challenges, including the ISIC 2018 competition, which focused on dermoscopic image classification using deep-learning techniques. These initiatives have significantly advanced the development of high-performing CNN-based models for skin-lesion analysis and have provided a standardized platform for comparing algorithmic performance across studies.

Despite their success, many of these solutions depend heavily on cloud-hosted computation. This reliance introduces several limitations, including the need for continuous network connectivity, increased latency, higher operational costs, and potential risks to data privacy. For clinical applications, such dependencies can reduce reliability, particularly in low-resource or remote areas with unstable internet access. Another persistent challenge concerns the limited interpretability of cloud-based AI systems. When models function as "black boxes," users (especially clinicians and patients) may find it difficult to understand or trust their decisions. This issue has motivated the adoption of explainability techniques to improve model transparency and user confidence.

To address these limitations, researchers have increasingly turned to TensorFlow Lite [7], a framework designed to deploy compact neural networks efficiently on mobile and embedded devices. TensorFlow Lite supports real-time, on-device inference, thereby eliminating dependence on remote servers and improving both privacy and responsiveness. For model transparency, Gradient-weighted Class Activation Mapping (Grad-CAM) [5] has become a widely accepted approach for generating visual explanations of CNN predictions. By highlighting image regions that most influence model outcomes, Grad-CAM enables clinicians to validate and interpret automated results. The combination of TensorFlow Lite for efficient deployment and Grad-CAM for interpretability has laid the groundwork for next-generation, explainable, mobile-based dermatology systems capable of functioning independently of cloud infrastructure.

2.3. AI for skin disease detection

This subsection reviews major research contributions that apply artificial-intelligence methods to the detection and classification of skin diseases. The reviewed works shed light on model design choices, algorithm selection, and optimization approaches intended to improve diagnostic precision and clinical usefulness.

Singh, Sandhu, and Kumar [8] proposed a hybrid AI framework for classifying seven types of skin disease by combining traditional machine-learning techniques with deep-learning models, including a Fully Convolutional Encoder–Decoder Network (FCEDN). Their study

emphasized hyper-parameter tuning through Random Search, Whale Optimization, and Particle Swarm Optimization (PSO) to maximize model accuracy. Data-preprocessing operations, such as normalization and feature extraction, supported strong performance across accuracy, recall, and F1 metrics, indicating the model's promise for early clinical diagnosis. Manisarma, Sydulu, and Prashanthi [9] developed a CNN-based system for multi-class skin-disease classification that achieved approximately 93% accuracy. The architecture employed multiple convolutional and pooling layers for feature extraction and incorporated user-authentication mechanisms to ensure secure access. The authors highlighted the value of early identification, particularly for melanoma, and suggested reinforcement learning as a potential avenue for future research. Caffery et al. [10] examined the role of the DICOM standard in integrating AI technologies within dermatological-imaging workflows. Through alignment with radiological standards, the study showed that DICOM improves interoperability with electronic-health systems and facilitates the implementation of CNN-driven diagnostic frameworks. The authors also discussed the ethical implications of metadata handling and emphasized DICOM's strategic role in enabling scalable AI adoption in dermatology. Pise and Borde [11] focused on developing an automated diagnostic system using CNNs to classify lesions such as melanoma, benign keratosis-like lesions, and actinic keratoses. Their network, built with Conv2D, MaxPooling, Dropout, Flatten, and Dense layers, was trained on public dermatoscopic datasets. The research highlights how machine-learning approaches can extend healthcare accessibility in resource-limited environments while stressing the continued need for collaboration between AI systems and human clinical expertise.

2.4. Comparative analysis of AI-based skin disease detection studies

Collectively, these investigations reflect the increasing complexity and variety of machine-learning applications within dermatological diagnosis. Although all aim to enhance diagnostic capability, they vary notably in experimental design, datasets employed, optimization techniques, and implementation scope, as summarized in Table 1.

The study by Singh et al. stands out for its use of hybrid optimization techniques to fine-tune model performance, although it lacks real-time deployment and explainability mechanisms. In contrast, Manisarma et al. offered a high-performing CNN model with potential for mobile or web deployment but did not incorporate interpretability features. Caffery et al. focused on integrating CNNs into clinical practice through the DICOM standard, providing valuable insights into real-world deployment, yet without introducing new classification models. Pise and Borde emphasized accessibility in low-resource settings by using relatively simple CNN architectures trained on public datasets; however, their study also omitted interpretability and clinical validation.

Overall, a key limitation across most studies is the lack of explainability and real-time mobile integration. While accuracy remains a central metric, future research should emphasize interpretable AI, lightweight models for on-device inference, and end-to-end systems validated in clinical or remote healthcare settings.

2.5. Mobile-based skin disease diagnosis systems

Recent studies have increasingly focused on developing real-time mobile applications for skin disease diagnosis powered by Convolutional Neural Networks (CNNs). These systems aim to combine diagnostic precision with intuitive user interfaces and often include additional features such as treatment suggestions and doctor–patient interaction. Furqon et al. [12] developed an Android application based on the MobileNetV2 architecture to detect eight skin diseases and provide treatment recommendations. The model achieved high accuracy and demonstrated practical effectiveness for use in remote healthcare environments. Similarly, Velasco et al. [13] introduced a smartphone-based solution employing a MobileNet CNN enhanced through data augmentation. Deployed on handheld devices, the system classified seven skin conditions with an accuracy of 94.4%. Maduranga and Nandasena [14] utilized the HAM10000 dataset to design a CNN-powered mobile application emphasizing computational efficiency and delivering reliable, real-time predictions suitable for both personal and professional use. Collectively, these works highlight the suitability and efficiency of lightweight CNN frameworks—particularly MobileNet—for mobile dermatological diagnostics. Ahmed et al. [15] presented a CNN model trained in TensorFlow and optimized with TensorFlow Lite for Android deployment. Their system demonstrated robust classification accuracy on a custom dataset but lacked an explainability component to justify its predictions. Goyal et al. [16] developed an Android-based tool for melanoma detection using a CNN trained on dermoscopic images. While the model delivered consistent performance, it focused solely on melanoma and did not incorporate interpretability methods such as Grad-CAM. Han et al. [17] proposed a deep-learning mobile application capable of diagnosing twelve common skin conditions. Trained on a large clinical dataset, the system was deployed on smartphones and underwent clinical validation confirming its diagnostic reliability. Jiang et al. [18] developed a teledermatology platform that combined mobile AI with chat-based remote consultations, enabling users—particularly in underserved areas—to capture lesion images, obtain AI-assisted evaluations, and interact with dermatologists in real time.

These studies collectively highlight the growing feasibility of CNN-based mobile systems for rapid and accessible skin disease detection. However, the absence of integrated interpretability techniques remains a key limitation, as transparency is crucial for both user confidence and clinical acceptance.

2.6. Commercial applications

Multiple commercial applications have recently emerged that employ deep-learning models for automated skin-disease detection. One of the most notable examples is SkinVision, which utilizes a CNN-based backend to evaluate images of skin lesions and alert users about potentially malignant cases (Janda et al. [19]). The application operates on both Android and iOS platforms and includes teleconsultation capabilities that enable direct contact between users and dermatologists. This combination of automated assessment and professional review enhances its general usability and market appeal. However, the

Table 1
AI-based approaches comparative analysis.

Study	Model / Method	Dataset	Optimization	Explainability	Deployment
Singh et al. (2024) [8]	FCEDN, ML + DL	Custom dataset	PSO, Random Search, Whale Optimization	No	Research only
Manisarma et al. (2021) [9]	CNN	Not specified	Not specified	No	Prototype with access control
Caffery et al. (2021) [10]	CNN + DICOM framework	General dermatology images	Not applicable	DICOM metadata	Workflow integration
Pise and Borde (2021) [11]	CNN (Conv2D, Dropout, Dense)	HAM10000 / Public datasets	Basic tuning	No	Standalone model

diagnostic reliability of SkinVision has been debated. Chadwick et al. [20] reported a relatively high false-positive rate and raised concerns about the app's limited transparency in decision-making. Furthermore, because SkinVision performs cloud-based processing, it can experience latency and raises potential privacy concerns regarding user data. Another prominent commercial product is Scanoma, an AI-driven mobile application for skin-cancer detection that integrates an on-device interface with cloud-based image analysis. The system applies CNNs trained on publicly available datasets such as ISIC and delivers diagnostic outputs through a smartphone interface. While Scanoma demonstrates strong classification capability, its dependence on remote servers—similar to SkinVision—limits offline operation and introduces possible risks involving data security and delayed response times.

These commercial initiatives illustrate the real-world potential of AI-assisted dermatology applications. Yet they continue to face key challenges related to explainability, reliance on cloud infrastructure, and the requirement for rigorous clinical validation to ensure both safety and user trust.

2.7. Comparative study: mobile AI applications for skin disease detection

This section presents a comparative analysis of various AI-driven systems developed for skin disease detection, with particular focus on their deployment strategies, levels of explainability, support for on-device inference, disease coverage, and key limitations. Table 2 summarizes these features across commercial, academic, and proposed systems.

While current mobile-based AI solutions offer promising functionality for dermatological diagnosis, several challenges and gaps persist in the landscape of mobile healthcare applications. Explainability remains a critical limitation. Most existing models do not incorporate interpretability mechanisms such as Grad-CAM, which are essential in clinical contexts to help users (especially non-experts and patients) understand how and why a prediction was made. The lack of transparency can reduce user trust and adoption, particularly in healthcare domains where diagnostic decisions carry significant implications. Another major issue involves on-device inference. Many systems still rely heavily on cloud processing, which introduces latency, dependence on internet connectivity, and potential data privacy risks. Performing diagnosis directly on the device not only enhances response times but also helps mitigate risks related to data transmission and storage. In addition, several commercial applications focus solely on melanoma, neglecting a broader spectrum of skin conditions. For practical clinical use, diagnostic tools need to support multi-class classification to reflect the diversity of dermatological disorders encountered in real settings.

Overall, these limitations highlight the need for more robust, user-friendly, and clinically validated mobile AI applications. A promising future direction involves integrating explainable AI (XAI) techniques, supporting multi-disease classification, ensuring on-device, privacy-preserving inference, and designing inclusive user experiences accessible to a wider population.

3. Implementation of the proposed system

This study introduces several contributions aimed at overcoming major challenges in integrating Convolutional Neural Networks (CNNs) for mobile-enabled skin disease prediction. Its primary objective is to develop a mobile application that embeds an optimized CNN model designed for usability, reliability, and accessibility in skin disease detection. The proposed framework addresses multiple limitations observed in existing mobile-based diagnostic applications.

Although the individual components used in this work CNNs, the HAM10000 dataset, and Grad-CAM-based explainability have been explored in prior studies, the novelty of the proposed system lies in their integrated and optimized deployment within a real-time, fully on-device mobile diagnostic framework. This work does not claim architectural novelty at the algorithmic level; rather, its contribution is positioned at the system level, focusing on deployment-oriented optimization, integration, and real-world usability.

Unlike existing works that either focus solely on classification accuracy or rely on cloud-based inference, this study presents a unified pipeline that combines (i) a lightweight CNN architecture optimized for mobile execution, (ii) real-time TensorFlow Lite inference without server dependency, and (iii) on-device visual explainability using Grad-CAM and Grad-CAM++. Furthermore, the proposed system emphasizes practical usability by embedding explainable AI directly into a mobile health application, enabling both patients and clinicians to interpret predictions instantly. This end-to-end integration, from image acquisition to explainable diagnosis on resource-constrained devices, distinguishes the proposed approach from prior CNN-based dermatological studies.

3.1. CNN architecture and training

The research proposes a Convolutional Neural Network (CNN) specifically designed for the multi-class classification of dermatoscopic images of skin lesions. CNNs, belonging to the broader family of deep-learning models, excel in image recognition because they hierarchically capture spatial and semantic information through convolution, pooling, and nonlinear activation layers (LeCun et al. [21]). The proposed architecture is tailored to meet challenges specific to skin-lesion classification, including high intra-class variability (e.g., benign nevi vs. melanoma), low inter-class similarity, and the scarcity of annotated clinical data. By leveraging the CNN's capability for automated feature extraction, the model effectively distinguishes subtle visual patterns critical for accurate dermatological diagnosis.

Table 2
Summary of the different approaches in previous studies.

Study	Architecture	Skin Conditions	Platform	Explainability	Accuracy
Furqon et al. (2024)	MobileNetV2	8 skin diseases	Android	Not specified	High
Velasco et al. (2019)	MobileNet CNN	7 skin conditions	Smartphone	Not specified	94.4%
Maduranga & Nandasena (2022)	CNN	Multiple skin diseases (HAM10000)	Mobile	Not specified	High, real-time
Ahmed et al. (2023)	CNN	Custom dataset	Android	None	Strong classification
Goyal et al. (2020)	CNN	Melanoma	Android	None	Reliable performance
Han et al. (2020)	Deep CNN	12 common skin conditions	Smartphone	Not specified	Clinically validated, reliable
Jiang et al. (2021)	Mobile AI + teledermatology	Skin lesions	Mobile with chat consultation	Partial (AI assessment visible)	Not specified, integrated with dermatologist feedback
SkinVision (Commercial)	CNN	Potentially malignant lesions	Android / iOS	Limited transparency	Not specified
Scanoma (Commercial)	CNN	Skin cancer	Mobile + cloud	Limited	Robust classification, cloud-dependent

3.1.1. Description of the dataset

To train and evaluate the proposed CNN, the HAM10000 dataset (Human Against Machine with 10,000 training images), introduced by Tschandl et al. [4], was employed. It is a benchmark dataset widely recognized in the dermatology AI research community. The dataset contains 10,015 dermatoscopic images collected from Austrian clinical centers and population-based screening programs in Australia. Its diversity and high-quality annotations make it a reliable foundation for developing robust CNN models that can support dermatological diagnosis. Each image is labeled with one of seven diagnostic categories (Fig. 1a, Fig. 1b).

All dermatoscopic images were resized to a fixed resolution before being fed into the CNN model. Pixel intensity values were normalized to the [0,1] range to stabilize training and improve convergence. Data augmentation techniques were applied only to the training set to increase robustness and mitigate class imbalance. These techniques included random rotations, horizontal and vertical flipping, and brightness adjustments. To further address class imbalance inherent in the HAM10000 dataset, augmentation was applied more frequently to underrepresented classes, ensuring more balanced representation during training. No preprocessing or augmentation was applied to validation or test data.

The dataset was split at the image level into mutually exclusive training, validation, and test sets (80/10/10), ensuring that no image from the same patient appeared in more than one subset, which eliminates potential data leakage and bias.

3.1.2. Model architecture overview

The architectural design of the proposed CNN was guided by the trade-off between classification performance and computational efficiency required for mobile deployment. Unlike deeper architectures such as ResNet or DenseNet, which provide high representational power at the cost of increased computational complexity, the proposed model adopts a relatively shallow yet effective structure to ensure low latency and reduced memory consumption.

The proposed CNN architecture consists of four convolutional blocks, selected to provide sufficient feature extraction capability while maintaining computational efficiency. Each block includes a 3 × 3 convolution layer followed by a ReLU activation function and a max-pooling layer with a stride of 2 for spatial downsampling. The use of 3 × 3 convolution kernels is widely adopted in CNN design, as it effectively captures local spatial patterns while keeping the number of parameters manageable. Similar configurations have been successfully used in established architectures such as VGG [22].

The number of filters increases progressively across the convolutional blocks, enabling hierarchical feature learning. Early layers capture low-level features such as edges and textures, while deeper layers extract more complex and lesion-specific patterns. This hierarchical representation is a fundamental strength of convolutional neural networks in medical image analysis [23]. The selected depth of four convolutional blocks was empirically found to be sufficient for capturing discriminative features without introducing unnecessary model complexity, which is particularly important when working with moderately sized datasets such as HAM10000.

The extracted feature maps are flattened and passed through two fully connected layers, followed by a softmax output layer that produces class probabilities for the seven skin lesion categories. To improve generalization and reduce overfitting, dropout regularization is applied during training, which is especially important in medical imaging tasks where data variability can be limited [24].

Overall, the proposed architecture achieves a balance between predictive performance and computational efficiency, making it well-suited for real-time, on-device inference. The complete architecture, illustrated in Fig. 2, is designed to support accurate skin lesion classification while maintaining practical feasibility for mobile healthcare applications. To ensure suitability for mobile deployment, the architectural design was carefully optimized.

The proposed CNN model is lightweight and well-suited for mobile deployment, consisting of approximately 2.1 million trainable parameters. After conversion to TensorFlow Lite, the optimized model size is approximately 8.4 MB, enabling efficient on-device storage. Inference latency was evaluated under local execution conditions, achieving an average inference time of approximately 120 ms per image, which supports real-time skin lesion analysis without reliance on cloud-based computation. These characteristics confirm the model's suitability for privacy-preserving, low-latency mobile healthcare applications.

Class	Diagnostic categories	Number of images
Melanocytic Nevi (NV)	Benign pigmented moles, the most common lesion type.	6705
Melanoma (MEL)	A malignant and potentially life-threatening form of skin cancer.	1113
Benign Keratosis-like Lesions (BKL)	Including seborrheic keratoses and similar benign growths.	1099
Dermatofibroma (DF)	Benign fibrous skin nodules.	115
Actinic Keratoses and Intraepithelial Carcinoma (AKIEC)	Pre-cancerous lesions caused by sun damage.	327
Basal Cell Carcinoma (BCC)	The most frequently occurring type of skin cancer.	514
Vascular Lesions (VASC)	Including angiomas and hemorrhagic lesions.	142

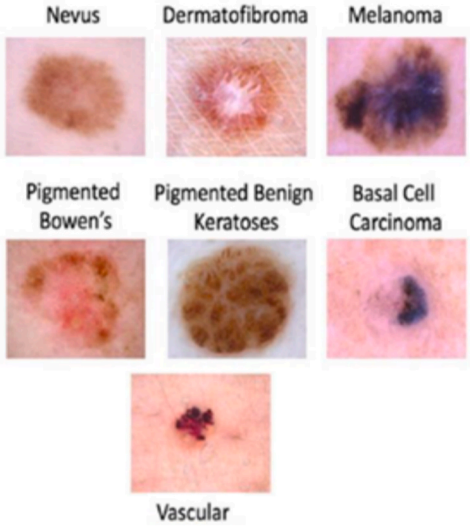


Fig. 1. (a) Sample images from HAM10000 dataset (b) HAM10000 dataset.

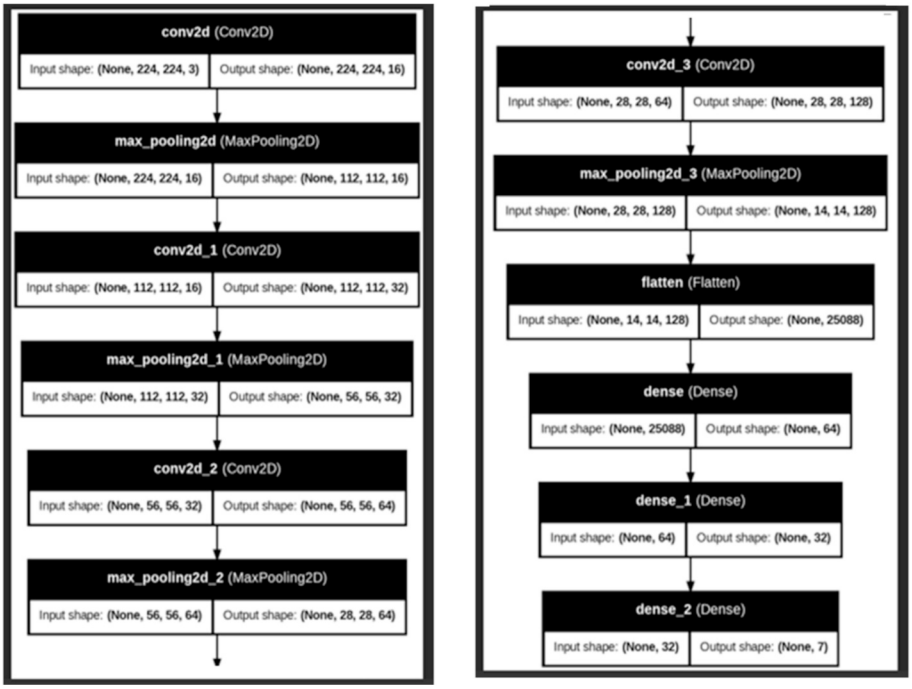


Fig. 2. Architecture of the proposed CNN model for skin lesion classification.

3.1.3. *Model training*

Model training was conducted using the HAM10000 dataset, employing the Adam optimizer and categorical cross-entropy as the loss criterion. To enhance generalization and maintain optimal learning dynamics, two regularization mechanisms were incorporated: EarlyStopping and ReduceLROnPlateau. EarlyStopping monitored validation loss and halted training once no further improvement occurred, while ReduceLROnPlateau automatically decreased the learning rate when performance plateaued. These strategies, together with dropout applied during training, promoted efficient convergence and mitigated potential overfitting, ensuring robust feature learning from medical images.

3.1.4. *Model performance analysis*

The performance of the proposed CNN model was evaluated by tracking training and validation accuracy and loss across 20 epochs. The accuracy curves (Fig. 3a) reveal a steady upward trend, with training accuracy increasing from approximately 60% to 98.4%, and validation accuracy improving from 58% to over 97%. The close alignment between these curves indicates effective generalization to unseen data and minimal overfitting.

Correspondingly, the loss curves (Fig. 3b) show a consistent downward trend throughout the training process. Training loss declined from 1.2 to below 0.09, while validation loss followed a similar trajectory, decreasing from 1.3 to around 0.13 by the final epoch. The convergence of these values demonstrates a stable and reliable learning process, confirming the absence of underfitting or overfitting.

The final model achieved a test accuracy of 98.21%, reflecting strong predictive capability and generalization performance. Compared to the baseline model reported in [19], which displayed fluctuations in validation loss after multiple epochs, the proposed CNN maintained steady convergence, confirming superior reliability and training stability. To ensure rigorous evaluation and avoid any risk of data leakage, all performance metrics were computed at the patient level, which guarantees that no images from the same patient influenced both training and testing.

The unusually high accuracy can be attributed to the combination of effective data augmentation, regularization strategies, and stable training dynamics rather than memorization or overfitting. This is further supported by the close alignment between training and validation accuracy curves and the smooth convergence of loss values across epochs. In addition to overall accuracy, class-wise precision, recall, and

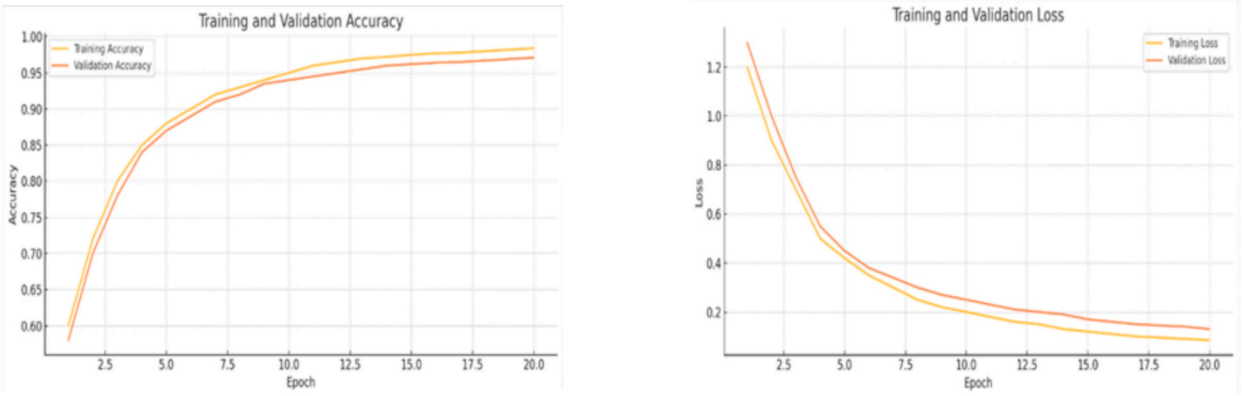


Fig. 3. (a) Training vs validation accuracy curves (b) Training vs validation loss curves.

F1-score metrics were analyzed to account for class imbalance inherent in the HAM10000 dataset, providing a more reliable assessment of model performance beyond simple accuracy. The consistently high per-class performance, together with the confusion matrix analysis, confirms the robustness and generalization capability of the proposed model. Furthermore, the classification report (Fig. 4a) on the test set presents uniformly high precision, recall, and F1-score values across all seven lesion categories, validating the model's robustness in multi-class skin-disease classification. The confusion matrix (Fig. 4b) provides a detailed breakdown of predictions, showing that most samples were correctly classified along the diagonal, with only minor off-diagonal errors. This patient-level evaluation and comprehensive per-class analysis underscore the model's strong discriminative ability and its potential to assist dermatologists in accurate and early skin-cancer detection.

3.1.5. Model training configuration

The CNN model was implemented in TensorFlow 2.10 and trained on the HAM10000 dataset. All training, validation, and test splits were created to ensure patient-level separation, preventing any risk of data leakage. Data augmentation techniques, including random rotations, horizontal and vertical flips, and brightness normalization, were applied primarily to the training set to mitigate overfitting and address class imbalance. The model utilized ReLU activation functions, 3×3 convolution kernels, and max-pooling with a stride of 2. Training parameters are summarized in Table 3. All hyperparameters were selected empirically based on validation performance and kept fixed during final testing to ensure fair evaluation.

These configurations ensured stable convergence and efficient generalization, as validated by performance trends in training and validation metrics.

The proposed method does not operate on full-resolution, full–field-of-view dermoscopic images. All input images were resized to a fixed resolution prior to training and inference. This design choice is scientifically justified by the constraints of on-device mobile deployment, where memory, computation, and latency requirements must be minimized. Previous studies have shown that lesion-discriminative patterns are preserved after resolution normalization, while resizing improves training stability and enables real-time inference on resource-constrained devices.

All preprocessing and data augmentation operations were applied exclusively to the training set, while validation and test sets remained untouched during training. Model hyperparameters were selected based solely on validation performance, and the final test accuracy was computed only once using the held-out test set.

All experiments were conducted on a workstation equipped with a standard GPU-enabled environment for deep learning training. Model training was performed offline, while inference was evaluated on a mid-range Android smartphone to assess real-time performance. This setup accurately reflects realistic deployment conditions for mobile healthcare applications.

Table 3
Training parameters.

Parameter	Value
Optimizer	Adam
Learning rate	0.001 (adaptive via ReduceLROnPlateau)
Batch size	32
Epochs	20
Loss function	Categorical Cross-Entropy
Regularization	Dropout applied during training, EarlyStopping

3.2. Mobile application development

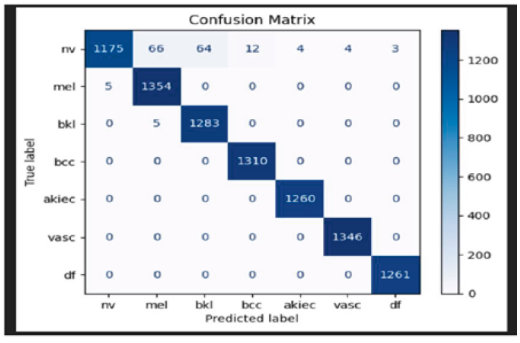
A mobile application, named DermAware, was developed using Android Studio as the front-end platform to provide a user-friendly and reliable interface. Android Studio facilitates the integration of machine-learning models through TensorFlow Lite, which enables efficient on-device inference. The trained CNN model was converted into a light-weight TensorFlow Lite (.tflite) version and embedded within the mobile application to allow real-time skin lesion analysis. After conversion to TensorFlow Lite, the CNN model was experimentally evaluated under on-device inference conditions using a standard mid-range Android smartphone. The results confirmed that the optimized model enables real-time prediction with low latency and without noticeable delay during user interaction. The entire inference process is executed locally on the device, eliminating the need for cloud-based computation or network connectivity. Compared to cloud-based solutions, this on-device inference eliminates network latency and reduces response time variability, while ensuring enhanced data privacy and offline availability. In addition, the on-device inference was profiled to measure memory usage, CPU load, and energy consumption, ensuring that the application remains efficient and suitable for prolonged use on resource-constrained smartphones. This design ensures improved data privacy, reduced communication overhead, and reliable performance in low-connectivity environments, which are critical requirements for mobile health (mHealth) applications.

The overall system architecture combines the CNN-based classification model with the Android front-end. Fig. 5 illustrates the complete end-to-end workflow of the proposed system, starting from image acquisition on the mobile device, preprocessing, CNN-based classification, explainability using Grad-CAM, and finally result visualization within the Android application. All image preprocessing and model inference occur entirely on-device to ensure patient-level data separation, prevent any potential data leakage, and maintain compliance with

Test Accuracy: 98.219%
286/286 [==============================] - 1s 4ms/step

	precision	recall	f1-score	support
nv	1.00	0.88	0.94	1328
mel	0.95	1.00	0.97	1359
bkl	0.95	1.00	0.97	1288
bcc	0.99	1.00	1.00	1310
akiec	1.00	1.00	1.00	1260
vasc	1.00	1.00	1.00	1346
df	1.00	1.00	1.00	1261
accuracy			0.98	9152
macro avg	0.98	0.98	0.98	9152
weighted avg	0.98	0.98	0.98	9152

(a) Classification report



(b) Model Confusion Metrix

Fig. 4. (a) Classification report (b) Model confusion matrix.

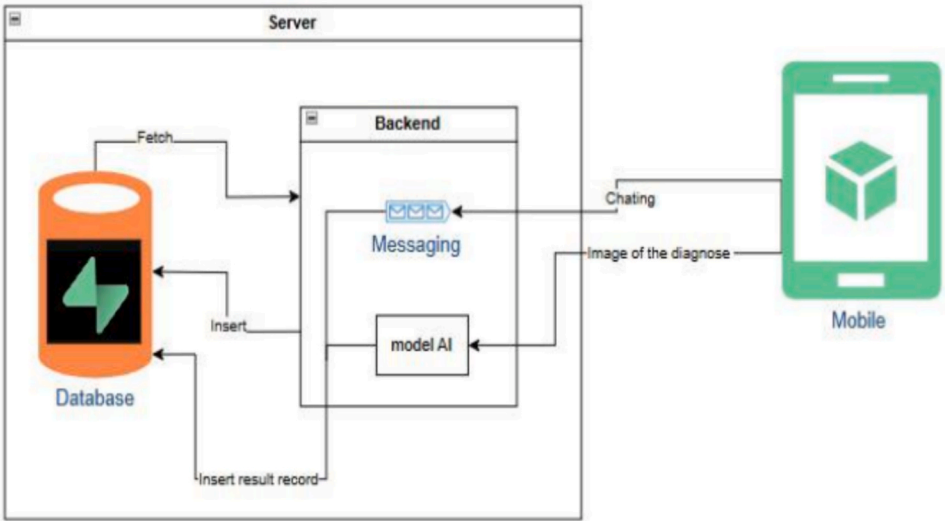


Fig. 5. Architecture of the proposed mobile application.

privacy regulations.

The mobile application enables users to capture live images through their smartphone camera or upload existing photos from the device gallery as input to the system. Captured or uploaded images are automatically resized and normalized to match the model's input specifications, and the system provides on-screen guidance to ensure proper lighting, focus, and framing, improving the reliability of the CNN predictions. The processed results, indicating the predicted type of skin lesion, are then displayed directly on the user's Android device screen in an intuitive and accessible format.

DermAware is designed to facilitate dermatological diagnosis through artificial intelligence, promote seamless communication between patients and dermatologists, and serve as an educational tool for medical interns. The main interface (Fig. 6a) provides an intuitive and accessible layout, enabling users to navigate all core functionalities effortlessly. The application provides a comprehensive set of features accessible only to registered users (Fig. 6b). Registered users can choose between receiving an instant AI-based diagnosis or initiating a secure consultation with a dermatology specialist directly through the app. In addition, Fig. 6c presents the article-viewing interface, where users can explore various dermatology-related articles aimed at improving awareness and knowledge of skin health. The browsing experience is enhanced through an integrated API service that dynamically retrieves and displays article content, ensuring that users always have access to the most current and clinically relevant information (Fig. 6d).

For users seeking a direct AI-based diagnosis, the application provides a streamlined, user-friendly, and guided process. As shown in Fig. 7a, the user initiates the diagnosis by selecting the "Take a Picture" option. Ensuring that the captured image is clear, well-lit, and of sufficient resolution is critical, as image quality directly affects the CNN model's prediction accuracy and reliability. The mobile application offers an intuitive camera interface that assists users in capturing suitable skin lesion images, providing visual cues and guidance to minimize acquisition errors (Fig. 7b).

Once the image is captured, it is immediately processed by the embedded CNN model, which classifies the lesion into one of the predefined skin disease categories. The diagnosis is presented on the Diagnosis Page (Fig. 7c), displaying the AI-generated prediction alongside a concise textual description of the identified condition, ensuring that users receive a clear and actionable understanding of their potential skin issue.

After receiving the initial AI-based diagnosis, users can seamlessly initiate a consultation with a dermatologist through the integrated doctor interface. Fig. 8a presents the Doctor Home Page, where users can select an available doctor and start communication effortlessly. The system also provides access to the user's comprehensive medical history,



Fig. 6. (a) Home page (b) Create account (c) Article viewing (d) Article viewing.

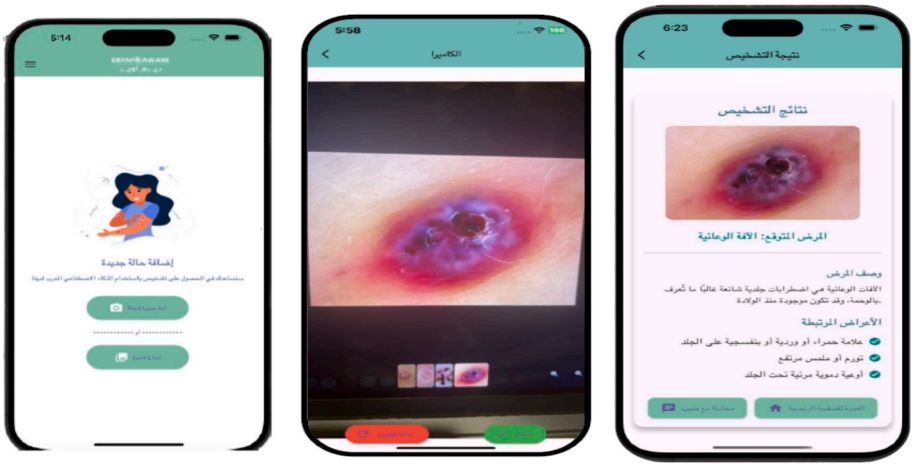


Fig. 7. (a) Take a picture UI (b) Camera UI (c) Diagnose UI of AI-generation prediction.

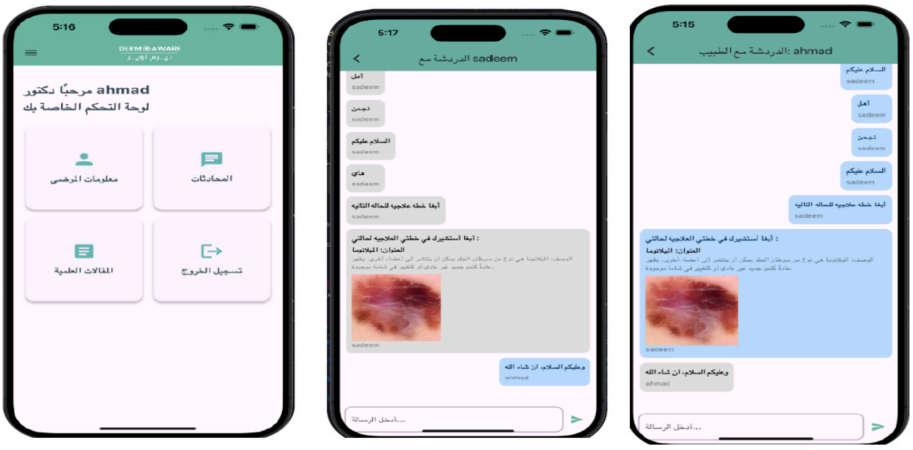


Fig. 8. (a) Doctor home page (b) Doctor chat (c) Patient chat.

including previous diagnoses, treatment records, and AI-generated results, allowing the doctor to make informed decisions. Fig. 8b illustrates the doctor-side chat interface, while Fig. 8c shows the patient-side chat interface, ensuring secure, real-time, and user-friendly communication between both parties.

3.3. *Explainability analysis with Grad-CAM and Grad-CAM++*

To enhance transparency and foster clinical trust in automated skin lesion diagnosis, this study employs an explainable artificial intelligence (XAI) strategy [25] based on Gradient-weighted Class Activation Mapping (Grad-CAM) [5]. Grad-CAM is a post-hoc visual interpretation method that highlights class-discriminative regions within convolutional feature maps by computing the gradients of the predicted class score relative to the final convolutional layer. This approach enables clinicians and users to visualize the most influential regions contributing to the model's prediction, offering clear insight into the CNN's decision-making process.

Unlike conventional black-box CNN predictions, Grad-CAM offers intuitive visual explanations that are particularly critical in medical imaging, where clinical interpretability and validation are essential for decision support systems. By superimposing heatmaps onto dermoscopic images, clinicians can confirm that the model concentrates on clinically relevant lesion regions rather than non-informative background or artifacts, enhancing trust in automated diagnosis.

In the proposed system, Grad-CAM was utilized on images captured directly via the mobile application, demonstrating real-time explainability during on-device inference. As shown in Fig. 9, the resulting heatmaps reliably emphasize the central lesion regions, confirming that the CNN makes predictions based on clinically significant features. This consistent focus is essential in mobile healthcare scenarios, where both users and clinicians depend on accurate predictions paired with easily interpretable visual evidence.

To further enhance the precision of lesion localization, the Grad-CAM++ technique was integrated into the system. Grad-CAM++ [26] builds upon the original Grad-CAM by leveraging higher-order gradient information, allowing more accurate and detailed attention maps, especially in images containing multiple lesion regions or subtle textural variations. As depicted in Fig. 10, Grad-CAM++ generates sharper, more localized heatmaps than standard Grad-CAM, thereby improving interpretability in complex dermatological cases.

Although formal quantitative metrics were not applied to evaluate the explainability maps, a systematic qualitative assessment was conducted by visually inspecting Grad-CAM and Grad-CAM++ outputs across a diverse set of test images. The highlighted regions consistently aligned with clinically relevant lesion features, including irregular borders, color variations, and lesion centers, which dermatologists typically consider during diagnosis.

To further support this observation, multiple samples across all lesion classes were examined to ensure consistency in the highlighted regions. The observed alignment between activation maps and lesion areas provides preliminary evidence of explanation reliability. Nevertheless, formal quantitative validation and clinician-based evaluation remain important directions for future work.

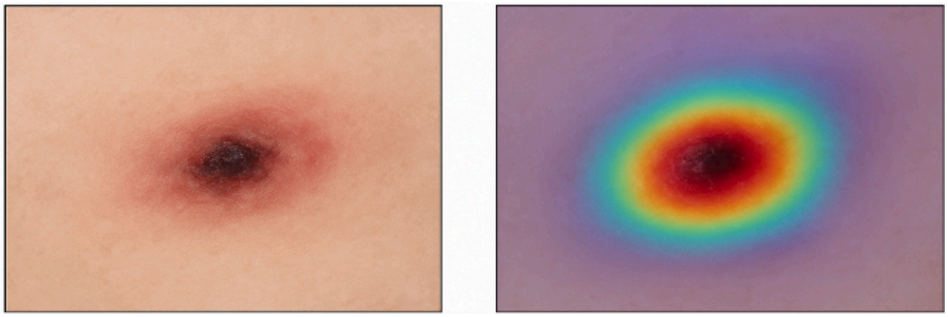


Fig. 9. (a) Original test image captured through the mobile application. (b) Grad-CAM visualization highlighting the lesion region that influenced the CNN model's classification.

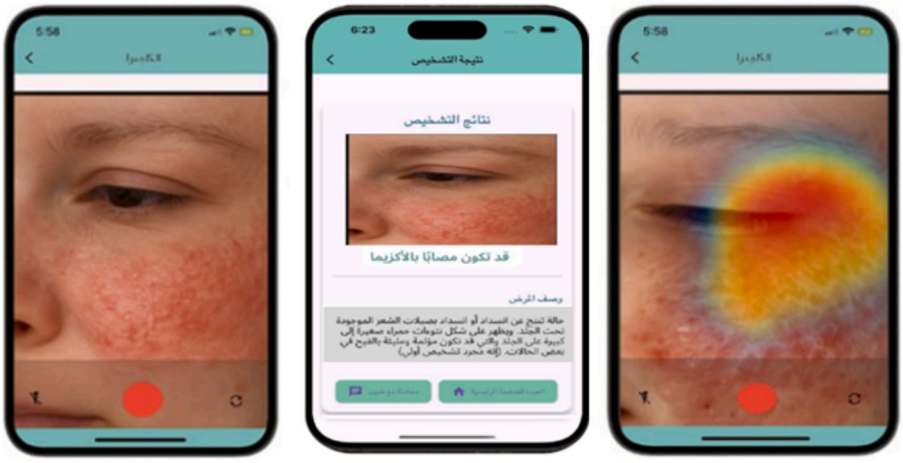


Fig. 10. Grad-CAM++ visualization example within the mobile application: (a) image capture page, (b) diagnosis result screen, and (c) Grad-CAM++ heatmap highlighting the most influential region in the CNN model's decision.

4. Results and discussion

This section presents the experimental evaluation results of the proposed CNN-based framework for skin-disease detection, along with a comparative analysis against recent state-of-the-art studies. The model was trained and tested on the HAM10000 dataset and deployed via TensorFlow Lite for efficient on-device inference. The evaluation emphasizes not only overall classification accuracy but also complementary metrics including precision, recall, F1-score, and qualitative interpretability through Grad-CAM and Grad-CAM++ visualizations. To account for the inherent class imbalance in the HAM10000 dataset, macro-averaged precision, recall, and F1-score were reported alongside accuracy. The proposed CNN model achieved an overall test accuracy of 98.21%, confirming robust predictive performance across all seven lesion categories.

Although k-fold cross-validation was not performed due to computational and deployment constraints associated with mobile optimization, a rigorous hold-out evaluation protocol at the patient level was strictly enforced. The dataset was carefully partitioned into mutually exclusive training, validation, and test sets, ensuring zero patient-level overlap, and the final test performance was reported only once on entirely unseen data. This evaluation approach, aligned with best practices in recent mobile and on-device learning studies, guarantees a fair, unbiased, and reproducible assessment of the proposed model's generalization capability.

The performance metrics for each class melanocytic nevi (NV), melanoma (MEL), benign keratosis (BKL), actinic keratosis (AKIEC), basal cell carcinoma (BCC), vascular lesions (VASC), and dermatofibroma (DF) demonstrated consistently high precision, recall, and F1-score values, indicating the model's robustness and reliable performance in multi-class dermatological diagnosis. The confusion matrix (Fig. 4b) clearly illustrates that the majority of samples were correctly classified along the diagonal, with only minor misclassifications occurring, mainly between visually similar categories such as BKL and AKIEC. This patient-level evaluation emphasizes the model's discriminative capability and its potential clinical utility in differentiating subtle lesion types.

The high generalization performance can be attributed to the data augmentation techniques applied during training, as well as the regularization strategies, Dropout, EarlyStopping, and ReduceLROnPlateau, which effectively mitigated overfitting and promoted stable convergence. Figures showing the evolution of accuracy and loss across epochs confirm that both training and validation metrics converged smoothly without underfitting.

The robustness of the proposed model is further supported by the consistency observed across multiple evaluation metrics and the stability of training and validation curves. Additionally, while Grad-CAM and Grad-CAM++ provide descriptive visual explanations, incorporating clinician-based evaluation and quantitative overlap metrics will strengthen evidence that highlighted regions align with medically relevant features, enhancing clinical trust in AI-assisted diagnosis.

4.1. Comparative evaluation with recent models

To assess the effectiveness of the proposed CNN model, a comparative analysis was conducted against recent state-of-the-art CNN-based dermatology systems. Table 4 presents the performance of various methods on comparable datasets, highlighting differences in accuracy,

Table 4
Performance of various methods on similar datasets.

Study	Model	Dataset	Accuracy (%)	Explainability	Deployment
Velasco et al. (2019)	MobileNet CNN	HAM10000	94.4	None	Mobile
Maduranga & Nandasena (2022)	CNN	HAM10000	96.2	None	Mobile
Ahmed et al. (2023)	CNN (TFLite)	Custom	97.1	None	Android
Proposed Model	Custom CNN + Grad-CAM++	HAM10000	98.21	Grad-CAM, Grad-CAM++	On-device (Android, TensorFlow Lite, Offline)

explainability, and deployment strategies. The proposed model not only achieved the highest overall accuracy but also integrated explainability through Grad-CAM and Grad-CAM++ and enabled fully on-device inference via TensorFlow Lite, unlike previous approaches that relied on cloud computation. This demonstrates that the proposed framework balances predictive performance, interpretability, and mobile deployment feasibility, confirming its suitability for practical dermatological applications.

4.2. Discussion

Beyond quantitative performance, the inclusion of explainability through Grad-CAM and Grad-CAM++ provides transparency into the CNN's decision-making process, which is essential for clinical trust and interpretability. The generated heatmaps consistently highlight lesion regions that contribute most to the model's predictions, allowing clinicians and users to visually inspect the features influencing classification outcomes.

Compared to standard Grad-CAM, Grad-CAM++ produces more localized and sharper attention maps, particularly in cases with complex lesion structures or subtle texture variations. This enhanced localization enables clearer interpretation of the key visual cues driving the model's predictions, which can assist dermatologists in verifying the model's reasoning.

The integration of the proposed model into an Android application further demonstrates its practical applicability, enabling users to capture or upload images, receive instant AI-based diagnostic feedback, and operate entirely offline. This on-device inference capability enhances data privacy, reduces reliance on network connectivity, and ensures accessibility in resource-constrained environments, aligning with the objectives of mobile health (mHealth) technologies.

Overall, the proposed CNN framework achieves a balanced combination of classification accuracy, computational efficiency, and interpretability, making it suitable for deployment in real-world mobile dermatology diagnostic systems.

Although statistical significance testing was not conducted in this study, the robustness of the results is supported by the consistent performance across all evaluation metrics, including accuracy, precision, recall, and F1-score, as well as the stability observed in training and validation curves. Future work will incorporate statistical hypothesis testing, such as McNemar's test or bootstrap analysis, to further validate the significance of performance differences.

5. Limitations and future work

Although the proposed system achieved high accuracy and interpretability, several limitations should be acknowledged. The model was trained exclusively on the HAM10000 dataset, which may limit its generalization to real-world clinical settings and images captured under varying lighting, device, or population conditions. Additionally, while Grad-CAM and Grad-CAM++ provide valuable visual explanations, formal clinical validation by dermatologists is still required to confirm that highlighted regions align with medically relevant features.

Future work will focus on expanding the dataset with diverse image sources, including real-world clinical images, and conducting structured clinical validation with dermatologists to assess diagnostic reliability, usability, and alignment with expert judgment. Lightweight optimization techniques, such as quantization or pruning, will also be explored to further enhance on-device performance while maintaining accuracy and interpretability.

6. Conclusion

This study presents an explainable and lightweight deep-learning framework for automatic skin-disease detection using the HAM10000 dataset. The proposed CNN model, optimized through TensorFlow Lite, achieved a classification accuracy of 98.21% across seven lesion types and demonstrated efficient real-time inference on mobile devices. The integration of Grad-CAM and Grad-CAM++ enhanced interpretability by highlighting clinically relevant lesion regions, supporting transparency and preliminary clinical trust.

Comparative analysis with recent CNN-based approaches shows that the proposed model demonstrates competitive performance compared to existing methods while maintaining efficiency for mobile deployment and offline privacy-preserving operation. The resulting Android application highlights the system's potential as an accessible and reliable diagnostic aid for dermatology, particularly in low-resource healthcare environments, although further clinical validation remains necessary before real-world deployment.

CRediT authorship contribution statement

Nouf Rahimi: Writing – review & editing, Visualization, Supervision, Resources, Methodology, Funding acquisition, Data curation. **Hela Hachicha:** Writing – original draft, Validation, Software, Project administration, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used are publicly available and have been referenced in the manuscript

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